CHAPTER 6

DISCRETE TIME FOURIER TRANSFORM

The techniques of continuous-time Fourier analysis developed in preceeding chapters are indispensable tools in analyzing and understanding the properties of continuous-time signals and systems.

In this chapter, we will try to parallel the same for discrete-time signals and systems.

6.1. RESPONSE OF DISCRETE TIME SYSTEMS TO COMPLEX EXPONENTIAL SEQUENCES

When the input to a discrete-time system with the impulse response $h[n]$ is

$$x[n] = z^n \tag{6.1.1}$$

where z is a complex number, the output will be

$$y[n] = z^n \sum_{k=-\infty}^{\infty} h[k]z^{-k} \quad , \tag{6.1.2}$$

as can be verified by the reader. This can be rewritten as

$$y[n] = H(z).z^n \tag{6.1.3}$$

where

$$H[z] = \sum_{k=-\infty}^{\infty} h[k]z^{-k} \quad . \tag{6.1.4}$$

From (6.1.3) we see that the response of a LTI discrete time system to an input of the type in (6.1.1) is of the same type. In other words z is the eigenfunction of a LTI discrete-time system, with $H(z)$ given by (6.1.4) the corresponding eigenvalue.

If the input to a LTI discrete time system is represented as a linear combination of complex exponentials, i.e.

$$x[n] = \sum_k a_k.z_k^n \quad , \tag{6.1.5}$$

then the output will be

$$y[n] = \sum_k a_k.H(z_k)z_k^n \quad . \tag{6.1.6}$$

We will, at this stage, restrict our discussion to complex exponentials of the form $e^{j\Omega n}$ (i.e. z^n with $|z| = 1$). The more general case will e treated later.

6.2. THE DISCRETE TIME FOURIER SERIES

We have seen earlier that the complex exponential sequence $e^{j(2\pi/N)n}$ where N is an integer, is periodic with period N. Furthermore, the set of all complex exponential sequences that are periodic with period N, i.e.

$$\Phi_k[n] = e^{jk(2\pi/N)n} \quad ; \quad k = 0, \mp 1, \mp 1, ... \tag{6.2.1}$$

are said to be harmonically related.

Let us note an important distinction between sets of harmonically related continuous-time and discrete-time signals: While all the harmonically related continuous-time signals satisfying the defining equation are distinct, there are only N distinct sequences satisfying (6.2.1).

A more general periodic sequence can be represented interms of harmonically related complex exponential sequences $\Phi_k[n]$ as

$$x[n] = \sum_k a_k \Phi_k[n] = \sum_k a_k . e^{jk(2\pi/N)n} \quad . \tag{6.2.2}$$

Since the sequences $\Phi_k[n]$ are distinct only over a range of N values of k, the summation of (6.2.2) need not include terms outside this range. This will be indicated in (6.2.2) as

$$x[n] = \sum_{k=k_0}^{k_0+N-1} a_k . e^{jk(2\pi/N)n} \quad , \tag{6.2.3}$$

where k_0 is an arbitrary integer. Equation (6.2.3), which consists of only a finite number of terms, will be referred to as the "discrete-time Fourier series", and a_k as the "Fourier series coefficients".

A natural question is about the existence and determination of such a Fourier series for a discrete time periodic signal. In other words, one would like to know when and how one can find a solution to the set of N equations

$$\begin{aligned} x[0] &= \sum_{k=k_0}^{k_0+N-1} a_k \\ x[1] &= \sum_{k=k_0}^{k_0+N-1} a_k . e^{jk(2\pi/N)} \\ &\vdots \\ x[N-1] &= \sum_{k=k_0}^{k_0+N-1} a_k . e^{jk(2\pi/N)(N-1)} \end{aligned} \quad . \tag{6.2.4}$$

It can be shown that these N equations are linearly independent and thus a_k, can be determined uniquely. Proof is left to the. reader. (Just write (6.2.4) in matrix form and identify a Vandermonde matrix!) Thus Fourier series for a finite sequence always exists and the N coefficients can be determined by solving (6.2.4).

Solution of (6.2.4) is not the only way to determine the Fourier series coefficients a_k. A closed form expression for a_k can be obtained from the equation

$$a_k = \frac{1}{N} \sum_{k=k_0}^{k_0+N-1} x[n] . e^{-jk(2\pi/N)n} \quad , \tag{6.2.5}$$

which can be derived by arguments paralleling those used for the continuous time case. This development will not be given here.

In summary, we have a "discrete-time Fourier series pair" repeated here for convenience

$$x[n] = \sum_{k=k_0}^{k_0+N-1} a_k . e^{jk(2\pi/N)n}$$
$$a_k = \frac{1}{N} \sum_{k=k_0}^{k_0+N-1} x[n] . e^{-jk(2\pi/N)n} \quad , \qquad (6.2.6)$$

The Fourier series coefficients a_k, often referred to as the "spectral coefficients" can be viewed as a sequence of numbers. Obviously this sequence is periodic with period N. However, only N successive values in this sequence will be used in the Fourier series representation.

The discrete-time Fourier series possesses a number of useful properties, many of which are direct counterparts of properties of continuous-time Fourier series. This detail will not be covered here.

We close this section by considering an example illustrating the use of discrete-time Fourier series.

Example 6.2.1

Suppose that a LTI system with the impulse response

$$h[n] = \alpha^n u[n] \quad , \quad -1 < \alpha < 1$$

is excited by

$x[n] = cos(2\pi n/N)$.

Then the Fourier series for the input is

$$x[n] = \frac{1}{2} e^{j2\pi n/N} + \frac{1}{2} e^{-j2\pi n/N}$$

and thus

$$y[n] = \frac{1}{2} H(2\pi/N) e^{j2\pi n/N} + \frac{1}{2} H(-2\pi/N) e^{-j2\pi n/N}$$

On the other hand

$$H(2\pi k/N) = \sum_{n=0}^{\infty} \alpha^n . e^{-j2\pi kn/N} = \sum_{n=0}^{\infty} \left(\alpha . e^{-j2\pi k/N}\right)^n$$
$$= \frac{1}{1 - \alpha . e^{-j2\pi k/N}}$$

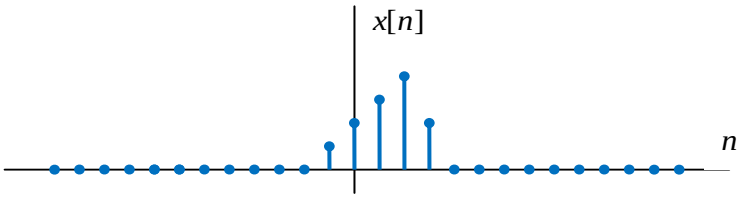
Therefore, for $N = 4$ *for example,*

$$y[n] = \frac{1}{2} \frac{e^{j2\pi n/N}}{1 - \alpha e^{-j2\pi n/N}} + \frac{1}{2} \frac{e^{-j2\pi n/N}}{1 - \alpha e^{j2\pi n/N}}$$
$$= \frac{1}{\sqrt{1+\alpha^2}} cos\left\{\frac{\pi n}{2} - tan^{-1}(\alpha)\right\}$$

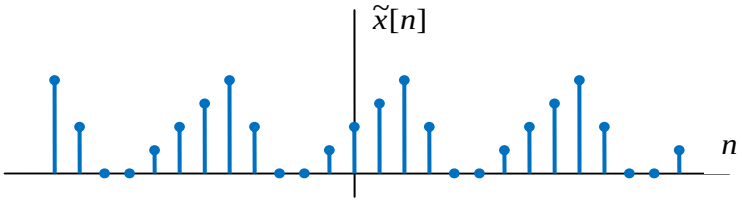
6.3. THE DISCRETE TIME FOURIER TRANSFORM

Extension of ideas presented in the preceeding section to aperiodic discrete-time sequences lead to the "discrete-time Fourier transform" in a manner parallel to that for continuous-time signals.

Let us consider an arbitrary sequence $x[n]$ of finite duration, as shown in Figure 6.3.la. This sequence satisfies $x[n] = 0$ for $|n| > N_1$ where N_1 is some integer. From this sequence, let us construct a periodic sequence $\tilde{x}[n]$ for which $x[n]$ is one period, as depicted in Figure 6.3.1b. Obviously as the period N of $x[n]$ is increased without bound, $\tilde{x}[n]$ approaches $x[n]$.



(a)



(b)

Figure 6.3.1. (a) Finite duration sequence $x[n]$; b) periodic sequence $\tilde{x}[n]$ equal to $x[n]$ over one period

The discrete-time Fourier series of $\tilde{x}[n]$ is

$$\tilde{x}[n] = \sum_{k=k_0}^{k_0+N-1} a_k . e^{jk(2\pi/N)n} \tag{6.3.1}$$

where

$$a_k = \frac{1}{N} \sum_{k=k_0}^{k_0+N-1} \tilde{x}[n] . e^{-jk(2\pi/N)n} \quad . \tag{6.3.2}$$

Since $\tilde{x}[n] = x[n]$ over a period that includes the interval $|n| > N$, the summation interval in (6.3.2) can be chosen to be this period so that (6.3.2) becomes.

$$a_k = \frac{1}{N} \sum_{k=-N_1}^{N_1} \tilde{x}[n] . e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{k=-\infty}^{\infty} x[n] . e^{-jk(2\pi/N)n} \quad . \tag{6.3.3}$$

If the envelope of Na_k is defined as

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] . e^{-j\Omega n}, \tag{6.3.4}$$

we observe that

$$a_k = \frac{1}{N} X(k\Omega_0) \tag{6.3.5}$$

where

$$\Omega_0 = 2\pi/N \tag{6.3.6}$$

is the sampling frequency. Thus (6.3.1) becomes

$$\tilde{x}[n] = \sum_{k=k_0}^{k_0+N-1} \frac{1}{N} X(k\Omega_0).\, e^{jk\Omega_0 n}$$
$$= \frac{1}{2\pi} \sum_{k=k_0}^{k_0+N-1} \Omega_0 X(k\Omega_0).\, e^{jk\Omega_0 n} \tag{6.3.1}$$

As was done for the continuous-time case, in the limit as N→∞, $\tilde{x}[n]$ becomes $x[n]$, and Ω_0 becomes arbitrarily small, transforming the summation in (6.3.7) into an integral, yielding

$$x[n] = \frac{1}{2\pi} \int_{\bar{\Omega}}^{\bar{\Omega}+2\pi} X(\Omega).\, e^{j\Omega n} d\Omega \tag{6.3.8}$$

where the integral can be taken over any interval of length 2π, i.e. $\bar{\Omega}$ is arbitrary.

As a result of the discussion above we have the discrete-time Fourier transform (DTFT) pair

$$x[n] = \frac{1}{2\pi} \int_{\bar{\Omega}}^{\bar{\Omega}+2\pi} X(\Omega).\, e^{j\Omega n} d\Omega$$
$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n].\, e^{-j\Omega n} \quad . \tag{6.3.9}$$

Heuristically speaking, first of equations (6.3.9) is a representation of the discrete-time sequence $x[n]$ as a linear combination of complex exponentials, infinitesimally close in frequency, each with amplitude $X(\Omega)\, {}^{d\Omega}/_{2\pi}$. The transform $X(\Omega)$ is often referred to as the "spectrum" of the sequence $x[n]$.

Although, we have based our discussion on a sequence of finite duration, (6.3.9) remains valid for a large class of sequences with infinite length. In this case question of convergence must be investigated. As it turns out, the conditions of existence and convergence are direct counterparts of those for continuous-time transform, and will not be repeated here. (For a detailed discussion of the convergence issues see one of the first two references). We will be content with stating that finite energy sequences defined by the property

$$\sum_{n=-\infty}^{\infty} [x[n]]^2 < \infty \tag{6.3.10}$$

or sequences that are absolutely summable, i.e.

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty \tag{6.3.11}$$

have discrete-time Fourier transforms.

One can easily observe that the continuous-time Fourier transform and the discrete-time Fourier transform (DTFT) are similar in many respects. The major differences are that, the DTFT $X(\Omega)$ is a periodic function of Ω and in the synthesis equation, the integral is taken over a finite interval. Both of these differences are results of the fact that discrete-time complex exponentials that differ in frequency by a multiple of 2π are identical.

Example 6.3.1.

Let us consider the sequence

$$x[n] = \alpha^n . u[n] \quad , \quad |\alpha| < 1 \ .$$

The DTFT for this sequence is

$$X(\Omega) = \sum_{n=-\infty}^{\infty} \alpha^n . u[n] . e^{-j\Omega n}$$
$$= \sum_{n=0}^{\infty} \alpha^n . e^{-j\Omega n} = \frac{1}{1 - \alpha . e^{-j\Omega}} \quad .$$

The amplitude and the phase of $X(\Omega)$ are shown in Figure 6.3.2 for $\alpha = 1/2$. Note the periodicity of both functions.

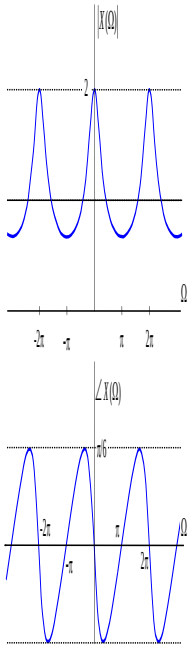


Figure 6.3.2. a) Magnitude; b) Phase plots for X(Ω) of Example 6.3.1,

A periodic sequence $x[n]$ never satisfies conditions like (6.3.10) or (6.3.11). But in case it is bounded, it possesses a discrete-time Fourier series representation and also a DTFT. As in the continuous time case, there are important relationships between the Fourier series representation for periodic sequences and their DTFT.

The Fourier series coefficients of a periodic sequence can be obtained as samples of the DTFT of one period of that sequence. More mathematically, if $x[n]$ is periodic with period N, and $x[n]$ is defined by

$$x[n] = \begin{cases} \tilde{x}[n] & M \le n \le M + N - 1 \\ 0 & \text{elsewhere} \end{cases} \qquad (6.3.12)$$

where M is an arbitrary integer, then it can be shown that

$$N a_k = X(2\pi k/N) \qquad (6.3.13)$$

where a_k are the Fourier series coefficients for $\tilde{x}[n]$ and $X(\Omega)$ is the DTFT of $x[n]$.

Example 6.3.2

Consider the periodic sample train

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN] \quad .$$

The aperiodic sequence

$$x_1[n] = \delta[n]$$

is equal to $x[n]$ over the period from 0 to N-1. The DTFT for this signal can be shown to be equal to

$$X_1(\Omega) = 1 \quad .$$

Therefore the Fourier series coefficients for $x[n]$ are

$$a_k = 1/N \quad .$$

as can be verified by direct calculation from (6.2.6).

As we have seen in Section 6.2 a periodic sequence can be expressed as a linear combination of harmonically related complex exponentials. The DTFT of the complex exponential

$$x[n] = e^{j\Omega_0 n} \tag{6.3.14}$$

is

$$X(\Omega) = \sum_{\mu=-\infty}^{\infty} 2\pi.\delta(\Omega - \Omega_0 - 2\pi\mu) \quad . \tag{6.3.14}$$

The reader can verify this last equation by placing it in the synthesis equation of (6.3.9) and noting that any interval of length 2π includes exactly one impulse. Therefore, when the general periodic sequence $x[n]$ is expressed as the linear combination of complex exponentials, i.e.

$$x[n] = \sum_{k=k_0}^{k_0+N-1} a_k.e^{jk(2\pi/N)n} \tag{6.3.15}$$

its Fourier transform will be

$$X[\Omega] = \sum_{k=k_0}^{k_0+N-1} a_k. \sum_{\mu=-\infty}^{\infty} 2\pi.\delta(\Omega - 2\pi k/N - 2\pi\mu) \quad . \tag{6.3.16}$$

Since the Fourier series coefficients are periodic, i.e. $a_k + N = a_k$, (6.3.16) can be rewritten as

$$X(\Omega) = \sum_{k=-\infty}^{\infty} 2\pi.a_k.\delta(\Omega - 2\pi k/N) \quad . \tag{6.3.17}$$

Example 6.3.3.

Let us consider the signal

$$x[n] = cos(\Omega_0 n) = \frac{1}{2}e^{j\Omega_0 n} + \frac{1}{2}e^{-j\Omega_0 n} \quad .$$

Therefore

$$X(\Omega) = \pi. \sum_{k=-\infty}^{\infty} \{\delta(\Omega - \Omega_0 - 2\pi\mu) + \delta(\Omega + \Omega_0 - 2\pi\mu)\}$$

which is nothing but a periodic repetition with period 2π, of the function

$$X_1(\Omega) = \pi.\{\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)\}$$

as shown in Figure 6.3-3.

6.4. PROPERTIES OF DISCRETE TIME FOURIER SERIES AND TRANSFORM

As with the continuous-time Fourier series and transform, discrete time Fourier series and transform enjoy several nice properties. There are many striking similarities with the continuous time case. Instead of discussing each one separately, they will be listed in the form of a table.

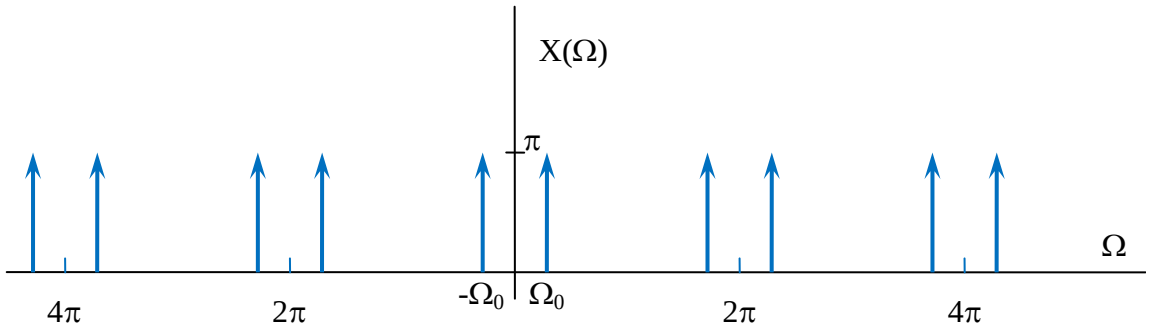


Figure 6.3.3. **DTFT** of $\cos \Omega_0 n$

Table 6.4.1. gives important properties of discrete-time Fourier transform, and some properties for discrete-time Fourier series are listed in Table 6.4.2. Using these properties and some basic transform pairs, more complicated transform pairs can be obtained. Some basic transform pairs are listed in Table 6.4.3. Whenever the sequence is periodic, the Fourier series coefficients are also shown. The reader is advised to derive the basic relations in the three tables.

Table 6.4.1. Properties of Discrete –time Fourier Transform

	Sequence	DTFT
	$x[n]$	$X(\Omega)$
	$y[n]$	$Y(\Omega)$
Linearity	$a.x[n]+b.y[n]$	$a.X(\Omega)+b.Y(\Omega)$
Time shifting	$x[n-n_0]$	$X(\Omega)e^{-j\Omega n}$
Frequency shifting	$x[n]e^{j\Omega n}$	$X(\Omega-\Omega_0)$
	$x^*[n]$	$X^*(-\Omega)$
	$x[-n]$	$X(-\Omega)$
Time scaling	$\tilde{x}[n]=\begin{cases} x[n/k] & \text{if n is a multiple of k} \\ 0 & \text{if n is not a multiple of k} \end{cases}$	$X(\mathrm{k}\Omega)$
Convolution	$x[n]*y[n]$	$X(\Omega).Y(\Omega)$
Modulation	$x[n].y[n]$	$\frac{1}{2\pi}\int_{2\pi} X(\theta)*Y(\Omega-\theta)d\theta$
Difference	$x[n]-x[n-1]$	$\left(1-e^{-j\Omega}\right)X(\Omega)$
Partial sum	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-e^{-j\Omega}}X(\Omega)+\pi X(0)\sum_{k=-\infty}^{\infty}\delta(\Omega-2\pi k)$
Frequency differentiation	$n.x[n]$	$j\frac{dX(\Omega)}{d\Omega}$
	$x[n]$ real	$X(\Omega)=X^*(-\Omega)$
	$x_e[n]=\mathcal{E}v\{x[n]\}$ $(x[n]$ real$)$	$\mathcal{R}e\{X(\Omega)\}$
	$x_o[n]=\mathcal{O}dd\{x[n]\}$ $(x[n]$ real$)$	$j\mathcal{I}m\{X(\Omega)\}$
Parseval's relation	$\sum_{n=-\infty}^{\infty}\lvert x[n]\rvert^2=\frac{1}{2\pi}\int_{\Omega_0}^{\Omega_0+2\pi}\lvert X(\Omega)\rvert^2 d\Omega$	

Table 6.4.2. Properties of Discrete –time Fourier Series

	Sequence	DTFS coefficients
	$\left.\begin{matrix} x[n] \\ y[n] \end{matrix}\right\}$ period N	$\left.\begin{matrix} a_k \\ b_k \end{matrix}\right\}$ period N
Linearity	$A.x[n] + B.y[n]$	$A.a_k + B.b_k$
Time shifting	$x[n - n_0]$	$a_k.e^{-jk\,(2\pi/N)n_0}$
Frequency shifting	$e^{jM\,(2\pi/N)}x[n]$	a_{k-M} (M: integer)
	$x^*[n]$	$a^*{}_{-k}$
	$x[-n]$	a_{-k}
Periodic convolution	$\sum_{\mu=\mu_0}^{\mu_0+N-1} x[\mu]\,y[n-\mu]$	$Na_k b_k$
Modulation	$x[n].y[n]$	$\sum_{\mu=\mu_0}^{\mu_{0+N-1}} a_\mu b_{k-\mu}$
Difference	$x[n] - x[n-1]$	$\left(1 - e^{-jk\,(2\pi/N)}\right)a_\mu$
Partial sum	$\sum_{k=-\infty}^{n} x[k]$ (finite valued and periodic only if a_0=0)	$\frac{a_k}{1 - e^{-j\Omega}}$
	$x[n]$ real	a_k=$a^*{}_{-k}$
	$x_e[n] = \mathcal{E}v\{x[n]\}$ ($x[n]$ real)	$\mathcal{R}e\{a_k\}$
	$x_o[n] = \mathcal{O}dd\{x[n]\}$ ($x[n]$ real)	$j\mathcal{I}m\{a_k\}$
Parseval's relation	$\frac{1}{N}\sum_{n=n_0}^{n_0+N-1} \lvert x[n]\rvert^2 = \sum_{k=k_0}^{k_0+N-1} \lvert a_k\rvert^2$	

Table 6.4. Basic Discrete –time Fourier Transform Pairs

Sequence, $x[n]$	DTFT $X(\Omega)$	DTFS coefficient, a_k (if periodic)
$\sum_{k=k_0}^{k_0+N-1} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta\left(\Omega - \frac{2\pi k}{N}\right)$	a_k
$e^{j\Omega_0 n}$	$2\pi \sum_{\mu=-\infty}^{\infty} \delta(\Omega - \Omega_0 - 2\pi\mu)$	$\begin{Bmatrix} 1 & k=m, m\mp N, m\mp 2N\ldots \\ 0 & \text{otherwise} \end{Bmatrix}$ for $\Omega_0/2\pi$ rational
$\cos \Omega_0 n$ $\Omega_0/2\pi = m/N$: rational	$\pi \sum_{\mu=-\infty}^{\infty} \{\delta(\Omega - \Omega_0 - 2\pi\mu) + \delta(\Omega + \Omega_0 - 2\pi\mu)\}$	$\begin{Bmatrix} \frac{1}{2} & k = \mp m, \mp m \mp N, \mp m \mp 2N, \ldots \\ 0 & \text{otherwise} \end{Bmatrix}$
$\sin \Omega_0 n$ $\Omega_0/2\pi = m/N$: rational	$\frac{\pi}{j} \sum_{\mu=-\infty}^{\infty} \{\delta(\Omega - \Omega_0 - 2\pi\mu) - \delta(\Omega + \Omega_0 - 2\pi\mu)\}$	$\begin{Bmatrix} \frac{1}{2j} & k = m, m \mp N, m \mp 2N, \ldots \\ \frac{-1}{2j} & k = -m, -m \mp N, -m \mp 2N, \ldots \\ 0 & \text{otherwise} \end{Bmatrix}$
1	$2\pi \sum_{\mu=-\infty}^{\infty} \delta(\Omega - 2\pi\mu)$	$a_k = \begin{Bmatrix} 1 & k=0, \mp N, \mp 2N\ldots\ldots \\ 0 & \text{otherwise} \end{Bmatrix}$
$\begin{Bmatrix} 1 & \lvert n\rvert \le N_1 \\ 0 & \text{else} \end{Bmatrix}$ and $x[n] = x[n+N]$	$2\pi \sum_{\mu=-\infty}^{\infty} a_k \delta(\Omega - \Omega_0 \mu)$ $\Omega_0 = 2\pi/N$	$\begin{Bmatrix} \frac{\sin[\Omega_0(N_1 + \frac{1}{2})]}{N \sin \Omega_0 n} & k\neq 0, \mp N, \mp 2N\ldots\ldots \\ \frac{2N_1 + 1}{N} & k=0, \mp N, \mp 2N\ldots\ldots \end{Bmatrix}$ $\Omega_0 = 2\pi/N$
$\sum_{\mu=-\infty}^{\infty} \delta[n - kN]$	$\frac{2\pi}{N} \sum_{\mu=-\infty}^{\infty} \delta(\Omega - 2\pi\mu/N)$	$a_k = {}^{1}/_{N}$
$\alpha^n u[n], \quad \lvert a\rvert < 1$	$\frac{1}{1 - \alpha . e^{-j\Omega}}$	-
$\begin{Bmatrix} 1 & \lvert n\rvert \le N_1 \\ 0 & \lvert n\rvert > N_1 \end{Bmatrix}$	$\frac{\sin[\Omega(N_1 + \frac{1}{2})]}{\sin[\Omega/2]}$	-
$\frac{\sin Wn}{n}$ $(0 < W < \pi)$	$\begin{Bmatrix} \pi & 0 \le \Omega \le W \\ 0 & W \le \Omega \le \pi \end{Bmatrix} \pi$ $(X(\Omega) = X(\Omega + 2\pi))$	-
$\delta[n]$	1	-
$u[n]$	$\frac{1}{1 - e^{-j\Omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\Omega - 2\pi k)$	-
$\delta[n - n_0]$	$e^{-j\Omega n_0}$	
$(n+1)\alpha^n u[n]$, $\lvert a\rvert < 1$	$\frac{1}{(1 - \alpha . e^{-j\Omega})^2}$	
$\frac{(n+r-1)!}{n!(r-1)!} \alpha^n u[n]$, $\lvert a\rvert < 1$	$\frac{1}{(1 - \alpha . e^{-j\Omega})^r}$	

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(*) of primary importance

PROBLEMS

Problem 1.

A LTI discrete-time system has an impulse response

$$h[n] = (1/2)^n u[n] .$$

Determine the Fourier series representation of the output y[n], to a periodic input x[n] with period N=4 where

$$x[n] = \begin{cases} 2 & n = 0 \\ -2 & n = 2 \\ 0 & n = 1 \text{ and } 3 \end{cases} .$$

Problem 2.

A discrete-time periodic signal given by

$$x[n] = \begin{cases} 1 & \text{n even} \\ 0 & \text{n odd} \end{cases}$$

drives a LTI system with a frequency response H(Ω) as shown in Figure P2. Find and sketch the output y[n].

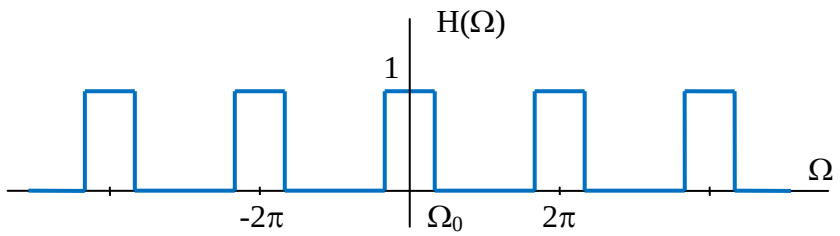


Figure P2

Problem 3.

Consider a continuous-time signal f(t). A discrete-time sequence is constructed from the samples of f(t) taken at a period of T, i.e. $f[n] = f(t)|_{t=nT} = f(nT)$.

On the other hand f(t) is impulse modulated to yield

$$g(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - nT) \quad .$$

Find a relationship between the discrete-time Fourier transform $F(\Omega)$ of f[n] and the Fourier transform $G(\omega)$ of g(t).

Problem 4.

Calculate

$$\sum_{k=-\infty}^{\infty} k.2^k = {}^1/_2 + {}^2/_{2^2} + {}^3/_{2^3} + \cdots \quad .$$

Problem 5.

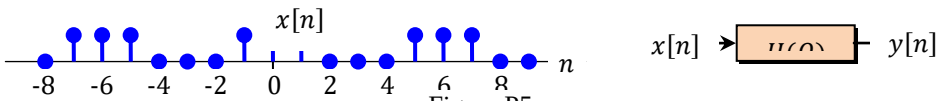


Figure P5

Consider the discrete-time sequence shown in the figure.

a) Determine the Fourier series expansion of $x[n]$, and sketch the spectrum (the Fourier series coefficients).

b) This sequence is applied to a linear, time-invariant system with a frequency response to yield an output $y[n]$. Determine and sketch $y[n]$.

Problem 6.

Let $X(\Omega)$ be the discrete-time Fourier transform of the sequence x[n] shown in Fig. (a). Determine the discrete-time Fourier transform of the sequence y[n] shown in Fig. (b), in terms of $X(\Omega)$. without calculating x[n] or y[n]. Derive any properties you use.

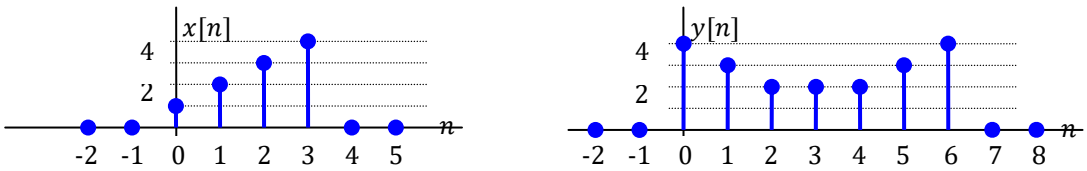


Figure P6

Problem 7.

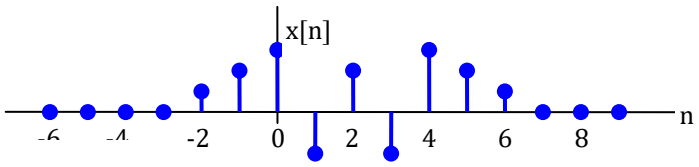


Figure P7

Let X(Ω) denote the discrete-time Fourier transform of the sequence x[n] shown in the figure. Without explicitly evaluating X(Ω), determine the following:

i) $X(\Omega)|_{\Omega=0}$;

ii) Phase of $X(\Omega)$;

iii) $\int_{-\pi}^{\pi} X(\Omega)d\Omega$;

iv) $\int_{-\pi}^{\pi} |X(\Omega)|^2 d\Omega$.

Problem 8.

If the discrete-time Fourier transform of the sequence x[n] is X(Ω), find the discrete-time Fourier transform of

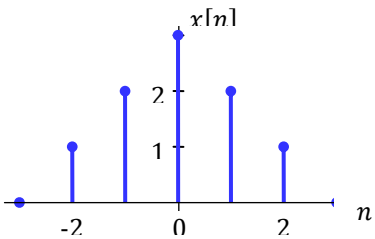
a) $x[n-1]$

b) $x[n+1]-x[n-1]$

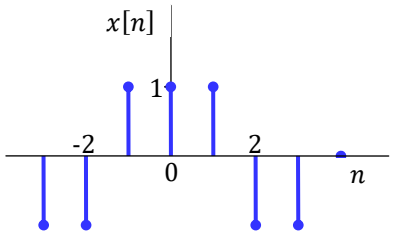
in terms of *X( )*.

Problem 9.

Determine the Fourier transforms of two signals shown in figure P??



(a)



(b)

Fig. P9

Problem 10.

Determine the Fourier transform of the sequence

$x[n] = (-1)^n \frac{1-e^{-T}}{T-jn2\pi}$.

Problem 11.

Determine the sequence x[n] , the Fourier transform of which is $X(\Omega) = e^{-\Omega}$, for $-\pi < \Omega < \pi$.

Problem 12.

In Chapter 4, we indicated that the continuous-time LTI system with impulse response

$$h(t) = \frac{W}{\pi}\,\mathrm{sinc}\left(\frac{Wt}{\pi}\right) = \frac{\sin(Wt)}{\pi t}$$

plays a very important role in LTI system analysis. The same is true of the discrete-time LTI system with impulse response

$$h[n] = \frac{W}{\pi}\,\mathrm{sinc}\left(\frac{Wn}{\pi}\right) = \frac{\sin(Wn)}{\pi n} \quad .$$

a) Determine and sketch the frequency response for the system with impulse response $h[n]$.

b) Consider the signal

$$x[n] = sin\left(\frac{\pi n}{8}\right) - 2cos\left(\frac{\pi n}{4}\right) \ .$$

Suppose that this signal is the input to a LTI systems with the following impulse responses. Determine the output in each case.

i) $h[n] \ = \frac{sin(\pi n/6)}{\pi n}$

ii) $h[n] = \frac{sin(\pi n/6)}{\pi n} + \frac{sin(\pi n/2)}{\pi n}$

iii) $h[n] \ = \frac{sin(\pi n/6)sin(\pi n/3)}{\pi^2 n^2}$

iv) $h[n] \ = \frac{sin(\pi n/6)sin(\pi n/3)}{\pi n}$

c) Consider a LTI system with unit sample response

$$h[n] = \frac{sin(\pi n/3)}{\pi n} \quad .$$

Determine the output for each of the following inputs:

i) $x[n]$ is the square wave depicted in the figure,

ii) $x[n] \ = \sum_{n=-\infty}^{\infty} \delta[n-8k] \ \ ,$

iii) $x[n] \ = \ (-1)^n$ times the square wave depicted in the figure

iv) $x[n] = \delta[n+1] + \delta[n-1]$.

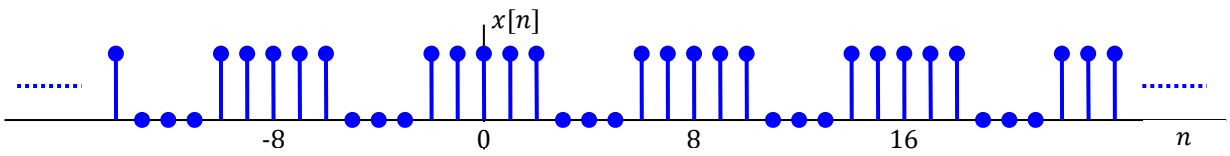


Figure P12